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EDUCATION

University of Chemistry and Technology, Prague

Ph.D. in Chemistry

Prague, Czech Republic

2022 – 2026

- **Supervisor:** Prof. Petr Slavíček
- **Dissertation:** *Computational Strategies for Nonadiabatic Molecular Dynamics*

University of Chemistry and Technology, Prague

Ing. in Physical Chemistry

Prague, Czech Republic

2020 – 2022

University of Chemistry and Technology, Prague

Bc. in Chemical Engineering

Prague, Czech Republic

2017 – 2020

EMPLOYMENT

University of Chemistry and Technology, Prague

Research/Teaching Position

Prague, Czech Republic

2022 – Present

- Research activities in the field of theoretical chemistry and photodynamics.
- Teaching undergraduate mathematics courses.

INTERNSHIPS

Christian-Albrechts-Universität zu Kiel

Research Internship (Kiel group)

Kiel, Germany

June 2026

- Group of Theoretical Spectroscopy and Heterogeneous Catalysis for solids, surfaces and condensed systems.
- **Supervisor:** Prof. Anna Hehn

PUBLICATIONS

1. **T. Jíra**, J. Janoš, and P. Slavíček.
Critical Assessment of Curvature-Driven Surface Hopping Algorithms.
Journal of Chemical Theory and Computation, 2025. DOI: [10.1021/acs.jctc.5c01176](https://doi.org/10.1021/acs.jctc.5c01176)
2. M. Buchowiecki and **T. Jíra**.
Spin-orbit induced charge transfer in $\text{H}^+/\text{D}^+/\text{T}^+-\text{N}(^4\text{S})$ collisions.
Journal of Quantitative Spectroscopy and Radiative Transfer, 2025. DOI: [10.1016/j.jqsrt.2025.109697](https://doi.org/10.1016/j.jqsrt.2025.109697)
3. **T. Jíra**, J. Janoš, and P. Slavíček.
Sensitivity Analysis in Photodynamics: How Does the Electronic Structure Control *cis*-Stilbene Photodynamics?
Journal of Chemical Theory and Computation, 2024. DOI: [10.1021/acs.jctc.4c01008](https://doi.org/10.1021/acs.jctc.4c01008)

SOFTWARE

Zinq

Code for quantum dynamics, mixed quantum-classical dynamics, and electronic structure methods.

 github.com/tjira/zinq